Nicolas Martin Guerra

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Education

PhD, Applied Mathematics Expected: May 2028
Cornell University | Ithaca, NY, USA
Advisor: Professor Yunan Yang

BS/MS, Applied Mathematics with concentration in Industrial Engineering June 2023
BM, Viola Performance
Northwestern University | Evanston, IL, USA
GPA: 3.93/4.00, *Cum Laude*

***Relevant Courses*:** Financial Engineering, Probability Theory, Mathematical Statistics, Linear Algebra, Partial Differential Equations, Statistical Learning for Data Analysis, Optimization Methods in Data Science, Machine Learning, Analysis of Algorithms, Service Engineering and Management

Research Experience

Graduate Research Assistant December 2023 – Present
Yunan Yang Research Group | Ithaca, NY

- Deriving theory for finding optimal training data distribution to maximize generalization of surrogate models in Scientific Machine Learning
- Developing theoretical foundations for Operator Learning, with applications in cancer detection and seismic imaging
- Designing machine learning architectures in PyTorch informed by Optimal Experimental Design for operator inference

Research Assistant January 2020 – June 2023
Ulmer Research Group | Evanston, IL

- Employed image filtering and dimension reduction algorithms in Python to detect exoplanets 163 light-years away
- Developed MATLAB programs for quantitative analysis of telescopic mirror roughness
- Managed over 100 GB of space-imaging data ensuring efficient data-accessing measures and organization

Undergraduate Researcher January 2022 – June 2022
Petia M. Vlahovska Research Group | Evanston, IL

- Performed eigenvalue stability analysis on nonlinear ODE systems to characterize chaotic dynamics of Quincke rotors
- Modeled simulations of the above system's behavior using MATLAB to gain a deeper understanding of its dynamics

Undergraduate Researcher June 2021 – September 2021
ISGC Summer Research Program | Evanston, IL

- Developed and automated Python programs to detect variable stellar sources and transient phenomena such as supernovae and gamma-ray bursts using image subtraction techniques
- Enhanced the speed and accuracy of original image processing procedures in the research lab by 40%
- Implemented efficient data handling pipelines for large-scale, remote Hubble datasets

Undergraduate Researcher June 2020 – September 2020
ISGC Summer Research Program | Evanston, IL

- Theorized methods to model imperfect mirrors using Zernike polynomials
- Developed MATLAB programs to model rays of light hitting imperfect telescopic mirrors using vector calculus

Honors and Awards

NDSEG Fellowship September 2025 – August 2028
Prestigious three-year fellowship awarded by the U.S. Department of Defense to outstanding PhD students in STEM fields, based solely on academic merit and research potential

Cornell Fellowship August 2023 – May 2024
Awarded to a select incoming PhD students in recognition of exceptional academic performance and research promise during undergraduate studies

Cornell Graduate School Dean's Scholar August 2023
Scholars who have demonstrated a strong commitment to academic excellence and advancing aspects of diversity, access, equity, inclusion, and belonging in the academy and other communities

Co-Winner of Northwestern's Applied Mathematics Outstanding Graduate June 2022
Recognition for outstanding achievement given by the advisors of the applied mathematics department

Illinois Space Grant Consortium (ISGC) Scholarship September 2021 – June 2022
Consortium part of NASA's National Space Grant College and Fellowship Program

Summer Undergraduate Research Grant June 2021 – July 2021
Northwestern University

ISGC Scholarship September 2020 – June 2021
Consortium part of NASA's National Space Grant College and Fellowship Program

Teaching Experience

Teaching Assistant August 2024 – December 2024
INFO 2950 Intro to Data Science, Cornell University | Ithaca, NY

- Led weekly discussion sessions and office hours to support learning in Python, SQL, and statistical modeling topics
- Designed and graded assessments for 250+ students, aligning evaluation with core data science concepts

Tutor Aide September 2022 – June 2023
America Reads @ McGaw YMCA | Evanston, IL

- Assisted elementary school children in reading, writing, and math to help them excel in school
- Facilitated students in completing and thoroughly understanding their homework assignments

Engineering Analysis IV Grader September 2022 – December 2022
Northwestern University | Evanston, IL

- Evaluated assignments and exams for a foundational differential equations course with over 100 students
- Provided constructive feedback on every assignment and addressed any uncertainties that students had

Instructor June 2019 – August 2019
iD Tech | Miami, FL

- Educated elementary and middle school students on programming fundamentals in Java and Lua
- Tailored my curriculum to align with each student's preferred learning style

Publications and Written Works

Guerra, N., Nelsen, N.H. and Yang, Y., 2025, May. Learning Where to Learn: Training Data Distribution Optimization for Scientific Machine Learning. *arXiv preprint arXiv:2505.21626*.

Ulmer, M.P., Dugard, J.H., Quispe, D., Buchholz, D.B., Stagon, S.P., Chung, Y.W., Cao, J., Kritikos, K., **Guerra, N.**, Stahl, M.T. and Shiri, R., 2022, August. A concept for a deployable normal incidence EUV mirror based on shape memory alloy sheets. In *Space Telescopes and Instrumentation 2022: Ultraviolet to Gamma Ray* (Vol. 12181, pp. 776-784). SPIE.

Ulmer, M.P., Jalilvand, M., Marks, N.A., Buchholz, D.B., Fujishima, B., **Guerra, N.**, Cao, J., Chung, Y.W., Baturalp, T.B., Coverstone, V.L. and Stagon, S.P., 2020, December. The prospects for applying magnetic smart materials combined with shape memory alloys to produce correctable and deployable space telescopes. In *Advances in Optical and Mechanical Technologies for Telescopes and Instrumentation IV* (Vol. 11451, pp. 393-404). SPIE.

Presentations

Learning Where to Learn.
Theory and algorithms for designing optimal training distributions to minimize out-of-distribution error.

- IMSI Data Assimilation and Inverse Problems for Digital Twins, University of Chicago, Oct 2025. (Oral)
- IMSI Data Assimilation and Inverse Problems for Digital Twins, University of Chicago, Oct 2025. (Poster)
- CAM Bill Sears Blitz, Cornell University, Sept 2025. (Oral)
- IMSI Statistical & Computational Challenges in Probabilistic Scientific ML, University of Chicago, June 2025. (Oral)
- IMSI Statistical & Computational Challenges in Probabilistic Scientific ML, University of Chicago, June 2025. (Poster)

Search for Transient Phenomena.
Detection of transient astronomical events using image subtraction and light-curve analysis.

- Annual ISGC Summer Research Presentations (Virtual), Aug 2021. (Oral)

Power Spectral Density.
Quantitative method for estimating mirror surface roughness via PSD analysis and its implications for imaging performance.

- Annual ISGC Summer Research Presentations (Virtual), Aug 2020. (Oral)

Skills

Professional

- MATLAB, Python (PyTorch, Numpy, Pandas), R/RStudio, SQL, Java, C, Docker, Github, AWS, Excel, GPU (CUDA), Machine Learning, statistical analysis, high-performance computing

Languages

- Spanish: Native fluency
- Vietnamese: Conversational fluency